

Worksheet 11

October 12, 2026

Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

1. Exponential function

Let

$$f(x) = 2^{x-1} - 3.$$

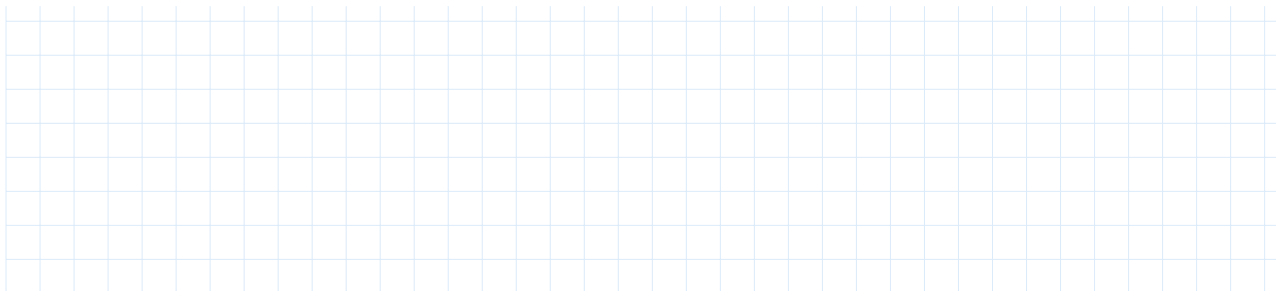
(a) Describe the horizontal and vertical shifts from the elementary exponential function.



(b) Give the horizontal asymptote.



(c) State the domain and range.



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2. A logarithmic equation

Solve

$$\log_3(x - 2) = 2.$$

(a) First state the restriction on x .

(b) Rewrite in exponential form, solve, and check the restriction.